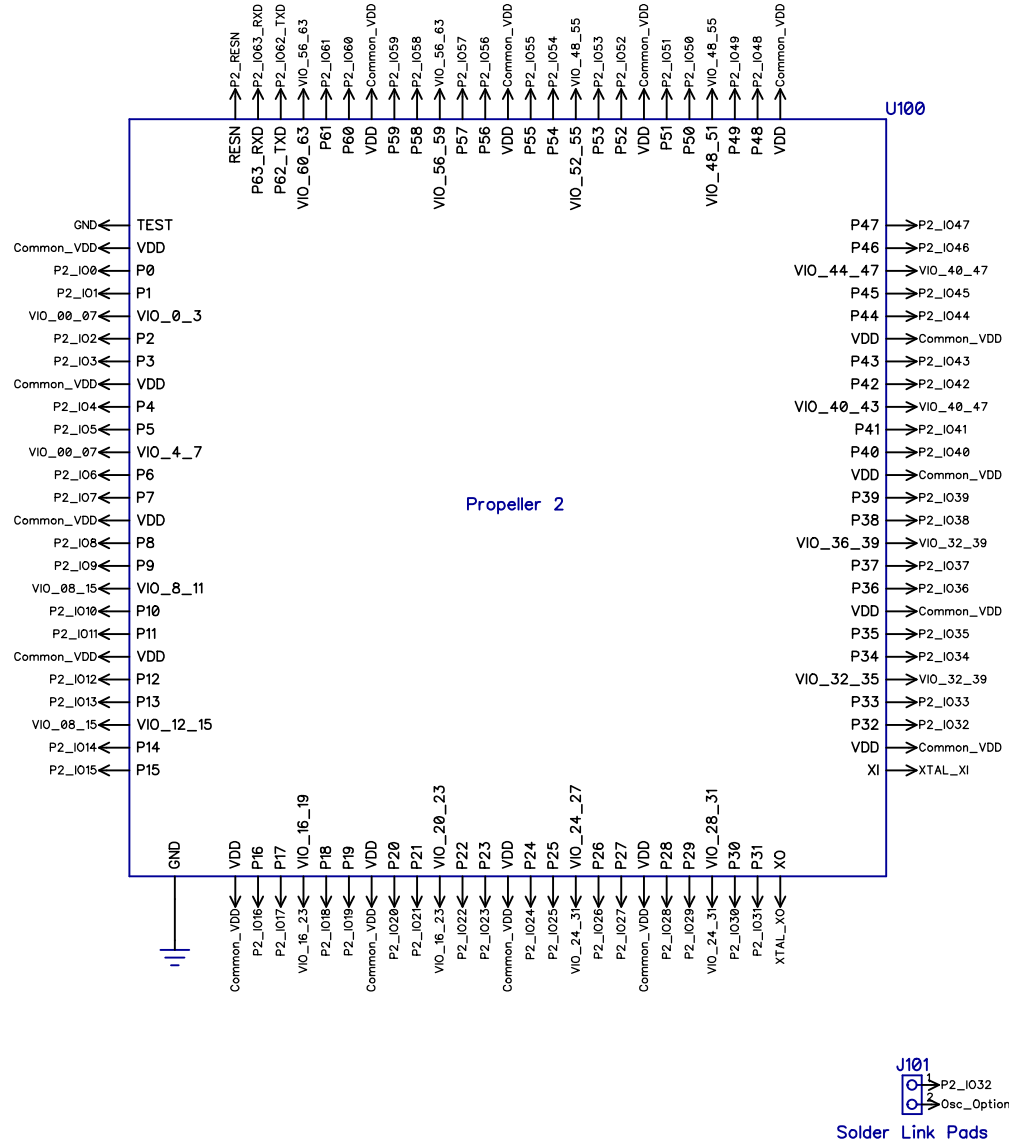
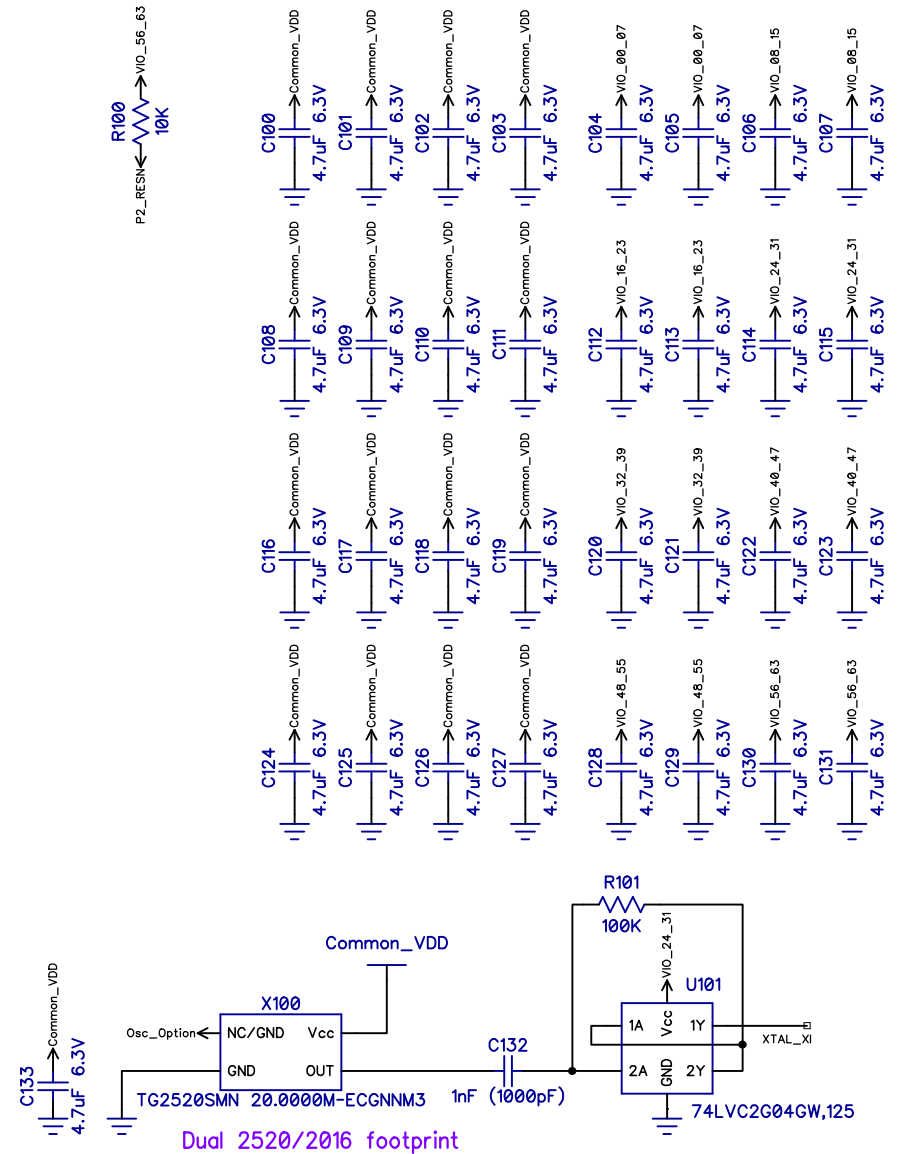


Microcontroller



VDD and VIO local decoupling



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<http://www.parallax.com/>

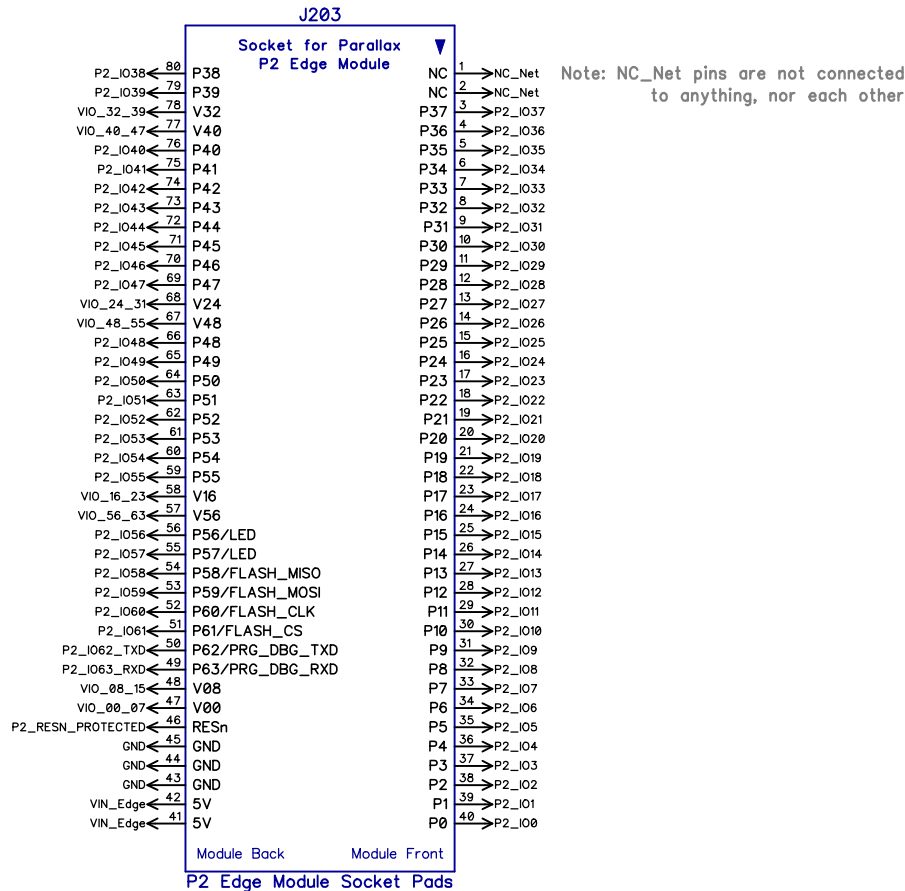
P2 Edge - Propeller 2 Module with Edge Connector

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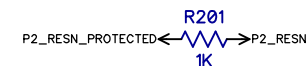
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P2 Edge Module, Card Edge Pads (80 way, 1.27mm pitch)



Reset pin protection



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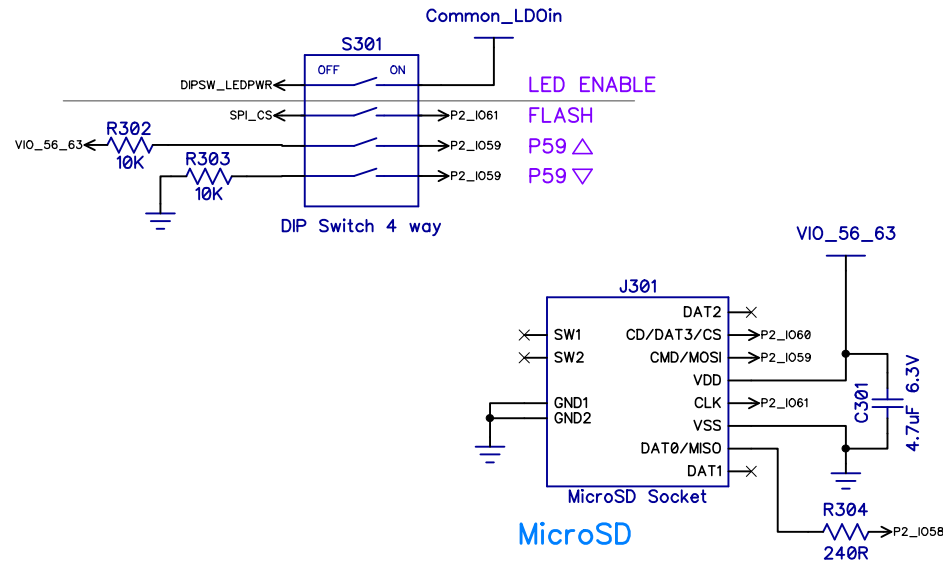
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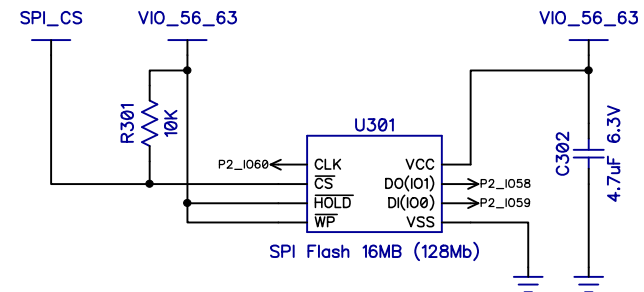
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Memory

Boot options and onboard LEDs enable switch



SPI Flash

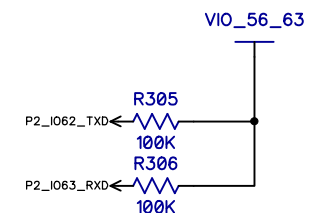


Boot Mode Selection	FLASH	P59 △	P59 ▽
Serial window of 60 seconds, default. (When SD card is NOT inserted)	OFF	OFF	OFF
Serial window of 60 seconds, overrides SPI Flash and SD card.	ON or OFF	ON	OFF
Serial window of 100 ms, then SPI flash. If SPI flash fails then serial window of 60 seconds.	ON	OFF	OFF
SPI flash only (fast boot), no serial window. If SPI flash fails then shutdown.	ON	OFF	ON
SD card with serial window on failure. If SD card fails then serial window of 60 seconds.	OFF	OFF	OFF
SD card only, no serial window. If SD card fails then shutdown.	OFF	OFF	ON

For reference: Boot Memory and Boot Serial (programming) pins

Boot Memory	P61 (output)	P60 (output)	P59 (output)	P58 (input)
SPI flash	CSn	CLK	DI	DO
SD card	CLK	CSn	DI	DO

Boot Serial	P63 (input)	P62 (output)
Serial	RX	TX



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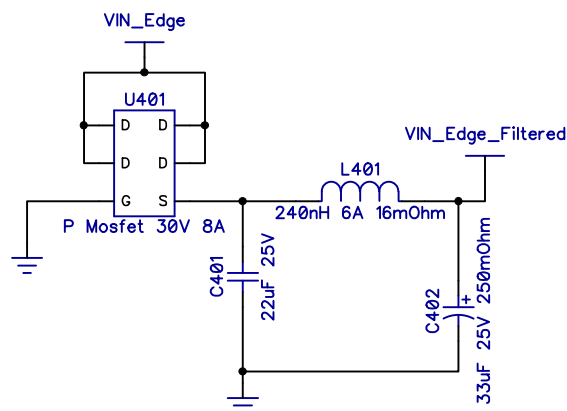
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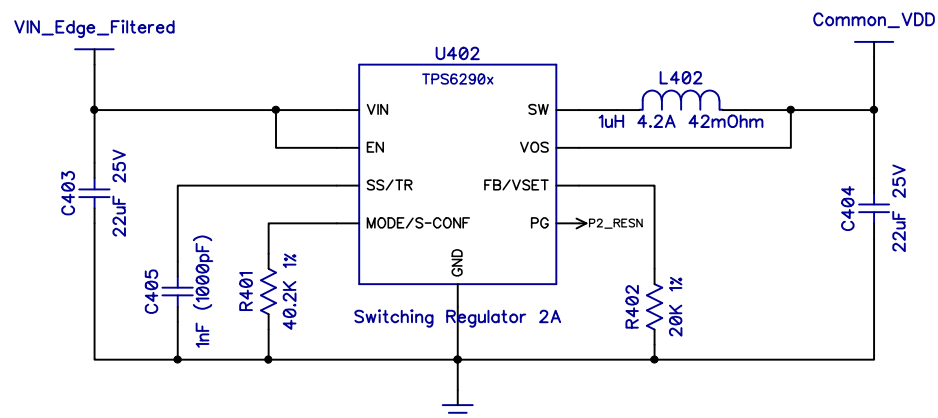
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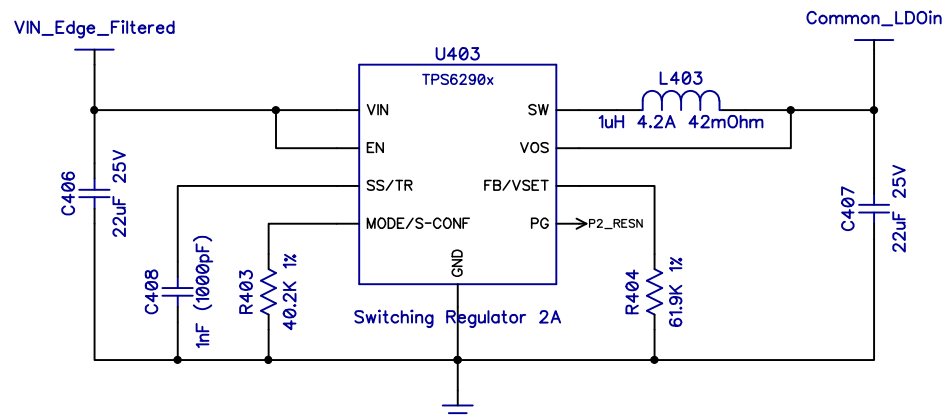
Reverse Polarity Protection and Input Filter



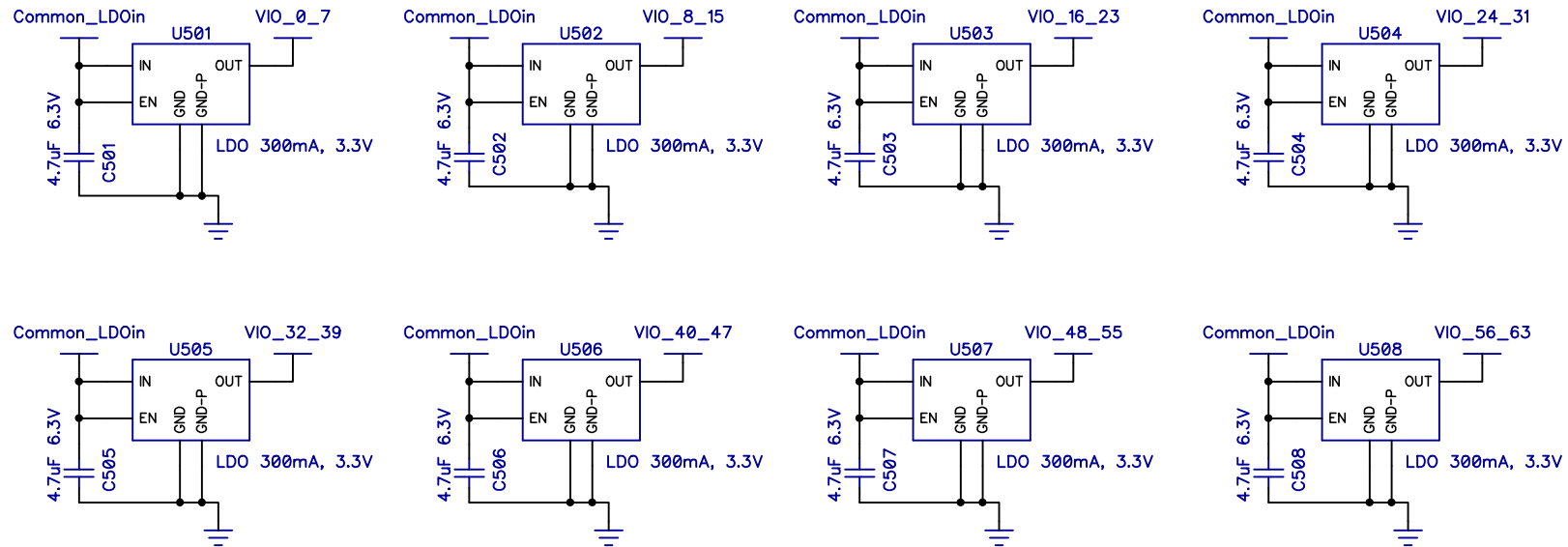
VDD Power Supply - Nominal 1.8V



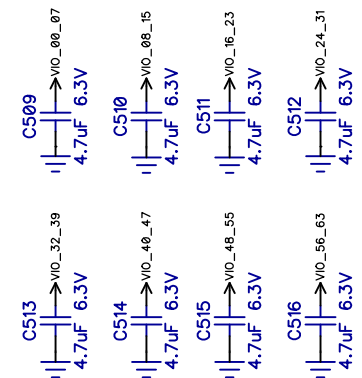
LDO Power Supply - Nominal 3.8V



Low noise VIO 3.3V local regulation



LDO output capacitors



These power sources ensure a fixed 3.3V low noise supply for each group of eight P2 smart pins.

A single high-current regulator could be used, but this design has some advantages, including:

- Less boardspace and lower part cost
- Thermal footprint spread over significantly larger area of copper
- Regulates directly to the power pins, removing power-path losses
- No requirement for local beads/caps for analog experiments, eliminates filtering losses
- Groups of IO's isolated from power supply cross-interference
- Limits power pin issues in each group from impacting others, including undervoltage, overvoltage and short-circuit protection

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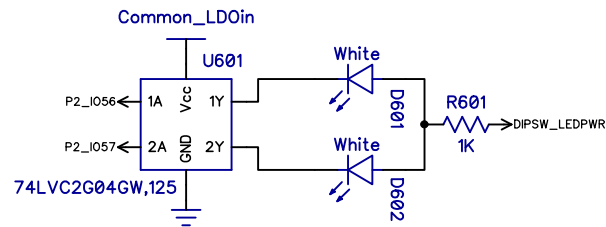
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LEDs



The LED's are wired such that they turn ON when the corresponding P2_IO pin asserts high.

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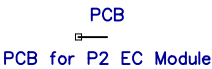
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Mounting holes



Mounting hole diameter 3.2mm
Orientation hole diameter 2.6mm

BOM only - mfg



RefDes	Quantity	Name	Manufacturer
C100, C101, C102, C103, C104, C105, C106, C107, C108, C109, C110, C111, C112, C113, C114, C115, C116, C117, C118, C119, C120, C121, C122, C123, C124, C125, C126, C127, C128, C129, C130, C131, C133, C301, C302, C501, C502, C503, C504, C505, C506, C507, C508, C509, C510, C511, C512, C513, C514, C515, C516	51	200-01049 : CAP-CER,4.7uF,X5R,6.3V,0402	TDK : C1005X5R0J475M050BC
C132, C405, C408	3	213-01014 : CAP-CER,1000pF,10%,16V,0402	Samsung : CL05B102K05NNNC
C401, C403, C404, C406, C407	5	200-11542 : CAP CER 22UF 25V X5R 0805	Samsung : CL21A226MAQNNNE
C402	1	202-10107 : CAP TANT POLY 33uF 25V 1411 250mOhm	AVX : TCNT336M025R0250
D601, D602	2	350-10034 : LED WHITE CLEAR 0603 R/A	Inolux : IN-S63AS5UW
J301	1	452-10033 : CON-MICRO SD,RA,SMT,LP	Molex : 473092651
L401	1	275-10022 : IND,240NH,6A,16MOHM	Samsung : CIGT252010LMR24MNE
L402, L403	2	275-10023 : IND,1uH,4.2A,42mOHM	Chilisin : BDCD003225121R0MS1
PCB	1	300-64001 : Raw PCB P2-EC Module	
R100, R301, R302, R303	4	150-11022 : RES,10K,5%,0402	Panasonic : ERJ-2GEJ103X
R101, R305, R306	3	150-11052 : Res,100k,0402	Yageo : RC0402JR-07100KL
R201, R601	2	173-10002 : RES,1K,5%,0402	Stackpole : RMCF0402JT1K00
R304	1	150-12023 : RES,240,5%,0402	Yageo : RC0402JR-07240RL
R401, R403	2	150-14711 : RES,40.2K,1%,0402	Yageo : RC0402FR-0740K2L
R402	1	172-16002 : RES,20K,1%,0402	Yageo : RC0402FR-0720KL
R404	1	172-16016 : RES,61.9K,1%,0402	Stackpole : RMCF0402FT61K9
S301	1	400-10100 : Slide Switch SPST DIP 4-way 1.27mm SMT	CTS Electro : 218-4LPSTJR
U100	1	P2X8C4M64P : P2X8C4M64P P2 multicore uC	Parallax : P2X8C4M64P
U101, U601	2	603-10039 : IC-INV,DUAL,74LVC2G04GW,SOT-23-6	NXP : 74LVC2G04GW,125
U301	1	604-10047 : IC FLASH 128M SPI 133MHZ 8SOIC	Winbond : W25Q128JVSIM TR
U401	1	500-10038 : MOSFET P-CH,30V,8A,TSOP6	Vishay : SI3417DV-T1-GE3
U402, U403	2	601-10342 : IC-REG,2A,9-VQFN,TPS62902	Texas Instruments : TPS62902
U501, U502, U503, U504, U505, U506, U507, U508	8	601-10339 : IC REG LDO CMOS 3.3V UDFN4	OnSemi : NCP114AMX330TCG
X100	1	252-10006 : Oscillator,20MHz,TCXO,1.8V,2520	EPSON : TG2520SMN 20.0000M-ECGNM3

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